

COMP0137 Computer Vision Cheatsheet

1 Probability Distributions

1.1 Bernoulli Distribution

$$\Pr(y) = \begin{cases} p, & y = 1, \\ 1 - p, & y = 0. \end{cases}$$

Notation: $y \sim \text{Bernoulli}(p)$.

1.2 Beta Distribution

$$\Pr(\lambda) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \lambda^{\alpha-1} (1 - \lambda)^{\beta-1}.$$

Key property:

$$\mathbb{E}[\lambda] = \frac{\alpha}{\alpha + \beta}.$$

1.3 Categorical Distribution

For K outcomes with parameters $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_K)$ (where $\sum_{k=1}^K \lambda_k = 1$),

$$\Pr(x = k) = \lambda_k.$$

For one-hot vector $\mathbf{x}$,

$$\Pr(\mathbf{x}) = \prod_{j=1}^K \lambda_j^{x_j}.$$

1.4 Dirichlet Distribution

$$\Pr(\lambda_1, \dots, \lambda_K) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \lambda_k^{\alpha_k - 1}.$$

1.5 Univariate Normal Distribution

$$\Pr(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Notation: $x \sim \mathcal{N}(\mu, \sigma^2)$.

1.6 Normal Inverse Gamma Distribution

$$\Pr(\mu, \sigma^2) = \sqrt{\frac{\gamma}{2\pi}} \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left(-\frac{2\beta + \gamma(\delta - \mu)^2}{2\sigma^2}\right).$$

1.7 Multivariate Normal Distribution

$$\Pr(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^D \det(\Sigma)}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

Notation: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

Covariance Structures:

- **Spherical:** $\Sigma = \sigma^2 I$.
- **Diagonal:** Σ is diagonal.
- **Full:** General covariance matrix with off-diagonals.

1.8 Normal Inverse Wishart Distribution

$$\Pr(\boldsymbol{\mu}, \Sigma) = \frac{\gamma^{D/2} |\Psi|^{\alpha/2} |\Sigma|^{-(\alpha+D+2)/2}}{(2\pi)^{D/2} 2^{\alpha\gamma/2} \Gamma_D(\alpha/2)} \times \exp\left\{-\frac{1}{2} [\text{Tr}(\Psi \Sigma^{-1}) + \gamma(\boldsymbol{\mu} - \boldsymbol{\delta})^\top \Sigma^{-1}(\boldsymbol{\mu} - \boldsymbol{\delta})]\right\}$$

1.9 Conjugate Distributions

- **Beta–Bernoulli:** Beta prior, Bernoulli likelihood $\rightarrow$ Beta posterior.
- **Dirichlet–Categorical:** Dirichlet prior, Categorical likelihood.
- **Normal–Inverse Gamma & Normal–Inverse Wishart:** Conjugate priors for Normal likelihoods.

2 Fitting Probability Models

2.1 Maximum Likelihood Estimation (MLE)

$$\mathcal{L}(\theta) = \Pr(\text{data} \mid \theta) = \prod_{i=1}^N \Pr(x_i \mid \theta).$$

Log-likelihood:

$$\log \mathcal{L}(\theta) = \sum_{i=1}^N \log \Pr(x_i \mid \theta).$$

Example (Normal):

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2.$$

2.2 Maximum A Posteriori (MAP)

$$\Pr(\theta \mid \text{data}) \propto \Pr(\text{data} \mid \theta) \Pr(\theta).$$

Example (Normal with Normal–Inverse Gamma prior):

$$\hat{\mu} = \frac{\sum_{i=1}^N x_i + \gamma \delta}{N + \gamma}, \quad \hat{\sigma}^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2 + 2\beta + \gamma(\delta - \mu)^2}{N + 3 + 2\alpha}.$$

2.3 Bayesian Approach

Posterior:

$$\Pr(\theta \mid \text{data}) = \frac{\Pr(\text{data} \mid \theta) \Pr(\theta)}{\Pr(\text{data})}.$$

Predictive Density:

$$\Pr(x^* \mid \text{data}) = \int \Pr(x^* \mid \theta) \Pr(\theta \mid \text{data}) d\theta.$$

3 The Normal Distribution

3.1 Univariate Normal

$$\Pr(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \sim \mathcal{N}(\mu, \sigma^2).$$

3.2 Multivariate Normal

$$\Pr(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^D \det(\Sigma)}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

Marginal Distributions:

If $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, then any subset $\mathbf{x}_A \sim \mathcal{N}(\boldsymbol{\mu}_A, \Sigma_{AA})$.

Conditional Distributions:

$$\Pr(\mathbf{x}_A \mid \mathbf{x}_B) \sim \mathcal{N}\left(\boldsymbol{\mu}_A + \Sigma_{AB} \Sigma_{BB}^{-1}(\mathbf{x}_B - \boldsymbol{\mu}_B), \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}\right).$$

Linear Transformations: For $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$,

$$\mathbf{y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^\top).$$

Product of Two Normals (Self-Conjugacy):

$$\mathcal{N}(\mu_1, \sigma_1^2) \times \mathcal{N}(\mu_2, \sigma_2^2) \propto \mathcal{N}(\mu_c, \sigma_c^2)$$

with

$$\mu_c = \frac{\mu_1/\sigma_1^2 + \mu_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}, \quad \sigma_c^2 = \frac{1}{1/\sigma_1^2 + 1/\sigma_2^2}.$$

Change of Variables:

$$\text{Norm}_x(\mathbf{A}\mathbf{y} + \mathbf{b}, \Sigma) = \kappa \text{Norm}_y(\mathbf{A}'\mathbf{x} + \mathbf{b}', \Sigma'),$$

where

$$\mathbf{A}' = \Sigma\mathbf{A}^\top(\mathbf{A}\Sigma\mathbf{A}^\top)^{-1}, \quad \mathbf{b}' = -\Sigma\mathbf{A}^\top(\mathbf{A}\Sigma\mathbf{A}^\top)^{-1}\mathbf{b}, \quad \Sigma' = (\mathbf{A}\Sigma\mathbf{A}^\top)^{-1}.$$

4 Components of the Solution

To solve an inference problem we need:

- **A Model:** A relationship between observed data $\mathbf{x}$ and world states $\mathbf{w}$ parameterized by $\boldsymbol{\theta}$.
- **A Learning Algorithm:** Fits $\boldsymbol{\theta}$ using training examples $(\mathbf{x}_i, \mathbf{w}_i)$.
- **An Inference Algorithm:** Computes $\Pr(\mathbf{w} \mid \mathbf{x})$ for new $\mathbf{x}$.

4.1 Discriminative Models: $\Pr(w \mid x)$

- Directly model $\Pr(w \mid x)$.
- **Learning:** Estimate parameters from (x, w) pairs.
- **Inference:** Evaluate $\Pr(w \mid x)$ on new data.

Examples:

(1) **Linear Regression (continuous w):**

$$\Pr(w \mid x) \sim \mathcal{N}(f_0 + f_1 x, \sigma^2)$$

(2) **Logistic Regression (binary $w \in \{0, 1\}$):**

$$\Pr(w \mid x) = \frac{1}{1 + \exp(-(f_0 + f_1 x))}.$$

4.2 Joint (Generative) Models: $\Pr(x, w)$

- Models the joint distribution of x and w .
- **Approach:**
 - (a) Concatenate x and w into $z = [x^\top, w^\top]^\top$.
 - (b) Model $\Pr(z)$ using, for example, a multivariate normal.
- **Inference:** Use Bayes' rule:

$$\Pr(w \mid x) = \frac{\Pr(x, w)}{\Pr(x)} \quad \text{with} \quad \Pr(x) = \int \Pr(x, w) dw.$$

4.3 Generative Models: $\Pr(x \mid w)$

- Model $\Pr(x \mid w)$ and combine with prior $\Pr(w)$.
- **Bayes' Rule:**

$$\Pr(w \mid x) = \frac{\Pr(x \mid w) \Pr(w)}{\Pr(x)}.$$

Example (Gaussian Likelihood):

$$\Pr(x \mid w) \sim \mathcal{N}(f_0 + f_1 w, \sigma^2).$$

5 Modeling Complex Densities

Techniques include:

- **Mixture of Gaussians (MoG):**

$$\Pr(x \mid \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k),$$

where latent variable h indicates component membership.

- **Student t-Distributions:**

$$\Pr(x) = \text{Stud}_x[\mu, \sigma^2, \nu] = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi\sigma^2} \Gamma(\frac{\nu}{2})} \times \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}.$$

with similar multivariate generalization.

- **Factor Analysis:**

$$\Pr(\mathbf{x}) = \text{Norm}_{\mathbf{x}}(\boldsymbol{\mu}, \boldsymbol{\Phi}\boldsymbol{\Phi}^\top + \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Phi}$ captures covariance structure in a low-dimensional subspace and $\boldsymbol{\Sigma}$ is diagonal.

6 EM Algorithm

6.1 Hidden (Latent) Variables

Introduce latent variables h :

$$\Pr(x) = \sum_h \Pr(x, h).$$

6.2 EM Algorithm Overview

Objective:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^I \log \int \Pr(x_i, h_i | \theta) dh_i.$$

Key Steps:

- **E-Step:** Compute posterior for latent variables:

$$q_i(h_i) = \Pr(h_i | x_i, \theta^{[t]}).$$

- **M-Step:** Update parameters:

$$\theta^{[t+1]} = \arg \max_{\theta} \sum_{i=1}^I \int q_i(h_i) \log \Pr(x_i, h_i | \theta) dh_i.$$

Lower Bound:

$$\mathcal{B}(\{q_i(h_i)\}, \theta) = \sum_{i=1}^I \int q_i(h_i) \log \frac{\Pr(x_i, h_i | \theta)}{q_i(h_i)} dh_i,$$

with Jensen's inequality ensuring $\mathcal{B} \leq \sum_{i=1}^I \log \int \Pr(x_i, h_i | \theta) dh_i$.

6.3 EM for Mixture of Gaussians (MoG)

E-Step: Compute responsibilities:

$$r_{ik} = \Pr(h_i = k | x_i, \theta^{[t]}) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}.$$

M-Step: Update parameters:

$$\begin{aligned} \pi_k^{[t+1]} &= \frac{1}{I} \sum_{i=1}^I r_{ik}, \\ \mu_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} x_i}{\sum_{i=1}^I r_{ik}}, \\ \Sigma_k^{[t+1]} &= \frac{\sum_{i=1}^I r_{ik} (x_i - \mu_k^{[t+1]})(x_i - \mu_k^{[t+1]})^\top}{\sum_{i=1}^I r_{ik}}. \end{aligned}$$

6.4 EM for Student t-Distributions

E-Step: For each x_i , compute:

$$\mathbb{E}[h_i] = \frac{\nu + D}{\nu + (x_i - \mu)^\top \Sigma^{-1} (x_i - \mu)},$$

and

$$\mathbb{E}[\log h_i] = \psi\left(\frac{\nu + D}{2}\right) - \log\left(\frac{\nu + (x_i - \mu)^\top \Sigma^{-1} (x_i - \mu)}{2}\right).$$

M-Step: Update:

$$\begin{aligned} \mu^{[t+1]} &= \frac{\sum_{i=1}^I \mathbb{E}[h_i] x_i}{\sum_{i=1}^I \mathbb{E}[h_i]}, \\ \Sigma^{[t+1]} &= \frac{\sum_{i=1}^I \mathbb{E}[h_i] (x_i - \mu^{[t+1]})(x_i - \mu^{[t+1]})^\top}{\sum_{i=1}^I \mathbb{E}[h_i]}. \end{aligned}$$

Degrees of freedom ν are updated via numerical methods.

6.5 EM for Factor Analysis

E-Step: Compute the posterior for latent factors:

$$\begin{aligned} q_i(\mathbf{h}_i) &= \Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) \\ &= \mathcal{N}\left((\Phi^\top \Sigma^{-1} \Phi + I)^{-1} \Phi^\top \Sigma^{-1} (\mathbf{x}_i - \mu), \right. \\ &\quad \left. (\Phi^\top \Sigma^{-1} \Phi + I)^{-1}\right). \end{aligned}$$

Compute expectations:

$$\mathbb{E}[\mathbf{h}_i] \quad \text{and} \quad \mathbb{E}[\mathbf{h}_i \mathbf{h}_i^\top].$$

M-Step: Update parameters:

$$\begin{aligned} \mu^{[t+1]} &= \frac{1}{I} \sum_{i=1}^I \mathbf{x}_i, \\ \Phi^{[t+1]} &= \left(\sum_{i=1}^I (\mathbf{x}_i - \mu) \mathbb{E}[\mathbf{h}_i]^\top \right) \left(\sum_{i=1}^I \mathbb{E}[\mathbf{h}_i \mathbf{h}_i^\top] \right)^{-1}, \\ \Sigma^{[t+1]} &= \frac{1}{I} \sum_{i=1}^I \text{diag}\left[(\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^\top - \Phi \mathbb{E}[\mathbf{h}_i] (\mathbf{x}_i - \mu)^\top\right]. \end{aligned}$$

6.6 Extended EM Details

General Objective:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^I \log \int \Pr(x_i, h_i | \theta) dh_i.$$

Lower Bound:

$$\mathcal{B}(\{q_i(h_i)\}, \theta) = \sum_{i=1}^I \int q_i(h_i) \log \frac{\Pr(x_i, h_i | \theta)}{q_i(h_i)} dh_i.$$

E-Step:

$$q_i(h_i) = \Pr(h_i | x_i, \theta^{[t]}) = \frac{\Pr(x_i, h_i | \theta^{[t]})}{\Pr(x_i | \theta^{[t]})}.$$

M-Step:

$$\theta^{[t+1]} = \arg \max_{\theta} \sum_{i=1}^I \int \Pr(h_i | x_i, \theta^{[t]}) \log \Pr(x_i, h_i | \theta) dh_i.$$

7 Linear Regression

7.1 Model Formulation

Assume the univariate target w is modeled by a normal distribution conditioned on the input x :

$$\Pr(w | x, \theta) = \mathcal{N}(w | \phi_0 + \phi^\top x, \sigma^2),$$

where:

- $\phi_0 + \phi^\top x$ is the mean (a linear function of x).
- σ^2 is the constant variance.

7.2 Compact Notation

To simplify notation:

- Append a constant 1 to every data vector:

$$x_i \leftarrow \begin{bmatrix} 1 \\ x_i \end{bmatrix}.$$

- Embed the offset into the parameter vector:

$$\phi \leftarrow \begin{bmatrix} \phi_0 \\ \phi \end{bmatrix}.$$

Thus, the model can be written as:

$$\Pr(w | x, \theta) = \mathcal{N}(w | \phi^\top x, \sigma^2).$$

7.3 Maximum Likelihood Estimation (MLE)

Maximize the log-likelihood:

$$\log \Pr(w | X, \theta) = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (w - X\phi)^\top (w - X\phi) \quad \text{so that}$$

Differentiating, the estimates obtained are:

$$\begin{aligned} \hat{\phi} &= (X^\top X)^{-1} X^\top w, \\ \hat{\sigma}^2 &= \frac{(w - X\hat{\phi})^\top (w - X\hat{\phi})}{N}. \end{aligned}$$

8 Bayesian Regression

8.1 Posterior Distribution

Assume the likelihood:

$$\Pr(w | X, \phi, \sigma^2) = \mathcal{N}(w | X\phi, \sigma^2 I),$$

and the prior:

$$\Pr(\phi) = \mathcal{N}(\phi | 0, \sigma_p^2 I).$$

Posterior via Bayes' rule:

$$\Pr(\phi | X, w) \propto \Pr(w | X, \phi) \Pr(\phi),$$

Closed-form posterior result:

$$\Pr(\phi | X, w) = \mathcal{N}\left(\phi | A^{-1} \frac{1}{\sigma^2} X^\top w, A^{-1}\right),$$

with

$$A = \frac{1}{\sigma^2} X^\top X + \frac{1}{\sigma_p^2} I.$$

8.2 Predictive Distribution

For a new data point x_* , the predictive distribution is obtained by marginalizing over ϕ :

$$\Pr(w_* | x_*, X, w) = \int \Pr(w_* | x_*, \phi) \Pr(\phi | X, w) d\phi.$$

This gives:

$$\Pr(w_* | x_*, X, w) = \mathcal{N}\left(w_* | \frac{1}{\sigma^2} x_*^\top A^{-1} X^\top w, x_*^\top A^{-1} x_* + \sigma^2\right).$$

In high dimensions, direct inversion of A can be costly. The matrix inversion lemma can be used to reduce complexity.

8.3 Fitting Variance

The noise variance σ^2 can be estimated by maximizing the marginal likelihood:

$$\Pr(w | X, \sigma^2) = \mathcal{N}(w | 0, \sigma_p^2 X X^\top + \sigma^2 I).$$

9 Non-Linear Regression

9.1 Model Formulation

Extend linear regression by transforming x into a higher-dimensional feature space z :

$$z_i = f(x_i),$$

$$\Pr(w_i | x_i, \theta) = \mathcal{N}(w_i | \phi^\top z_i, \sigma^2).$$

9.2 Examples of Basis Functions

- **Polynomial Basis:**

$$z_i = \begin{bmatrix} 1 \\ x_i \\ x_i^2 \\ \vdots \\ x_i^K \end{bmatrix}.$$

- **Radial Basis Functions (RBFs):**

$$z_i = \begin{bmatrix} \exp\left(-\frac{(x_i - c_1)^2}{\lambda}\right) \\ \exp\left(-\frac{(x_i - c_2)^2}{\lambda}\right) \\ \vdots \end{bmatrix}.$$

- **Arc Tan Functions:**

$$z_i = \begin{bmatrix} \arctan(\lambda(x_i - \alpha_1)) \\ \arctan(\lambda(x_i - \alpha_2)) \\ \vdots \end{bmatrix}.$$

9.3 MLE for Non-Linear Regression

With design matrix Z of transformed features:

$$\hat{\phi} = (Z^\top Z)^{-1} Z^\top w,$$

and variance:

$$\hat{\sigma}^2 = \frac{(w - Z\hat{\phi})^\top (w - Z\hat{\phi})}{N}.$$

10 Kernel Regression and Gaussian Processes

10.1 The Kernel Trick

Instead of computing z_i explicitly, use a kernel function:

$$K(x_i, x_j) = z_i^\top z_j.$$

This enables working implicitly in high-dimensional feature spaces.

10.2 Example Kernels

- **Linear Kernel:** $K(x_i, x_j) = x_i^\top x_j$.
- **Polynomial Kernel (degree p):**

$$K(x_i, x_j) = (x_i^\top x_j + 1)^p.$$

- **RBF (Gaussian) Kernel:**

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{\lambda^2}\right).$$

10.3 Gaussian Processes (GPs)

A Gaussian process places a prior over functions:

$$w(x) \sim \mathcal{GP}(m(x), k(x, x')),$$

where $m(x)$ is the mean function (often zero) and $k(x, x')$ is the kernel defining the covariance.

10.4 Gaussian Process Regression

For a new x_* , the predictive distribution is:

$$\Pr(w_* | x_*, X, w) = \mathcal{N}(\mu_*, \sigma_*^2),$$

with

$$\mu_* = K(x_*, X) \left[K(X, X) + \sigma^2 I \right]^{-1} w,$$

$$\sigma_*^2 = K(x_*, x_*) - K(x_*, X) \left[K(X, X) + \sigma^2 I \right]^{-1} K(X, x_*).$$

11 Sparse Linear Regression

11.1 Model Formulation

To encourage sparsity, impose a sparsity-inducing prior on ϕ :

$$\Pr(\phi) = \prod_{d=1}^D \text{Student-}t(\phi_d | 0, 1, \nu).$$

This penalty drives many ϕ_d towards zero, effectively selecting only the most informative features.

11.2 Latent Variable Reformulation

Express the t -distribution as a marginal using hidden variables:

$$\Pr(\phi) = \int \prod_{d=1}^D \mathcal{N}\left(\phi_d | 0, \frac{1}{h_d}\right) \text{Gamma}\left(h_d | \frac{\nu}{2}, \frac{\nu}{2}\right) dh_d.$$

Large h_d drive the corresponding ϕ_d to zero, thereby enforcing sparsity.

12 Dual Linear Regression

12.1 Dual Representation

Express the parameter vector as:

$$\phi = X\psi,$$

so that the model becomes:

$$\Pr(w | X, \theta) = \mathcal{N}(w | X^\top X \psi, \sigma^2 I).$$

This representation depends only on the dot products $X^\top X$ and is suitable for kernelization.

12.2 MLE and Bayesian Formulation (Dual)

- **MLE:**

$$\hat{\psi} = (X^\top X)^{-1} X^\top w,$$

$$\hat{\sigma}^2 = \frac{(w - X^\top X \hat{\psi})^\top (w - X^\top X \hat{\psi})}{N}.$$

- **Bayesian Dual Regression:** With the prior:

$$\Pr(\psi) = \mathcal{N}(\psi | 0, \sigma_p^2 I),$$

the posterior is:

$$\Pr(\psi | X, w, \sigma^2) = \mathcal{N}\left(\psi | A^{-1} \frac{1}{\sigma^2} X^\top w, A^{-1}\right),$$

where

$$A = \frac{1}{\sigma^2} X^\top X + \frac{1}{\sigma_p^2} I.$$

- **Predictive Distribution:** For a new x_* :

$$\Pr(w_* | x_*, X, w) = \mathcal{N}\left(w_* | \frac{1}{\sigma^2} x_*^\top A^{-1} X^\top w, x_*^\top A^{-1} x_* + \sigma^2\right).$$

13 Relevance Vector Machines

RVMs combine the dual representation with sparsity priors to yield models that depend on a small subset of data points (relevance vectors).

13.1 Model Formulation

$$\Pr(w \mid X, \theta) = \mathcal{N}(w \mid X^\top X \psi, \sigma^2 I),$$

with a prior on dual variables:

$$\Pr(\psi) = \prod_{i=1}^N \text{Student-}t(\psi_i \mid 0, 1, \nu).$$

During training, hidden variables h_i are updated so that many of the ψ_i go to zero, leaving only the relevance vectors.

13.2 Kernelization

Since the formulation depends only on dot products $X^\top X$, the kernel trick can be employed to work in high-dimensional feature spaces.

14 Logistic Regression

14.1 Model Formulation

Logistic regression models the posterior probability $\Pr(w \mid x)$ for binary outcomes ($w \in \{0, 1\}$). It assumes:

$$\Pr(w \mid \phi, x) = \text{Bern}_w[\sigma(a)],$$

where:

- $a = \phi_0 + \phi^\top x$ is the linear activation.
- $\sigma(a) = \frac{1}{1 + e^{-a}}$ is the sigmoid function.

The full model is expressed as:

$$\Pr(w \mid \phi, x) = (\sigma(a))^w (1 - \sigma(a))^{1-w}.$$

14.2 Likelihood and Log-Likelihood

Likelihood over Dataset: For data $\{x_i, w_i\}_{i=1}^I$,

$$\Pr(w \mid X, \phi) = \prod_{i=1}^I \sigma(a_i)^{w_i} (1 - \sigma(a_i))^{1-w_i}, \quad a_i = \phi^\top x_i.$$

Log-Likelihood:

$$\mathcal{L}(\phi) = \sum_{i=1}^I [w_i \log \sigma(a_i) + (1 - w_i) \log (1 - \sigma(a_i))].$$

Gradient of the Log-Likelihood:

$$\frac{\partial \mathcal{L}(\phi)}{\partial \phi} = \sum_{i=1}^I [\sigma(a_i) - w_i] x_i.$$

This gradient forms the basis for iterative gradient-based optimization.

14.3 Optimization Techniques

1. **Gradient Descent:**

$$\phi^{[t+1]} = \phi^{[t]} - \eta \nabla \mathcal{L}(\phi^{[t]}),$$

where η is the learning rate.

2. **Line Search:** Select the step size λ along the gradient:

$$\lambda = \arg \min_{\lambda} f(\phi^{[t]} + \lambda \nabla \mathcal{L}(\phi^{[t]})).$$

3. **Newton's Method:** Using the Hessian H (second derivatives):

$$\phi^{[t+1]} = \phi^{[t]} - H^{-1} \nabla \mathcal{L}(\phi^{[t]}),$$

where

$$H = \frac{\partial^2 \mathcal{L}(\phi)}{\partial \phi^2}.$$

4. **Finite Differences:** Numerically approximate the gradient:

$$\frac{\partial f}{\partial \phi_j} \approx \frac{f(\phi + \epsilon e_j) - f(\phi)}{\epsilon},$$

with e_j the unit vector and ϵ a small step size.

14.4 Decision Boundary

The decision boundary is where $\Pr(w = 1 \mid x) = 0.5$, corresponding to:

$$\phi_0 + \phi^\top x = 0.$$

For non-linear extensions, the boundary can be non-linear in the original feature space.

15 Bayesian Logistic Regression

15.1 Likelihood with Prior

• **Likelihood:**

$$\Pr(w \mid X, \phi) = \prod_{i=1}^I \sigma(a_i)^{w_i} (1 - \sigma(a_i))^{1-w_i}.$$

• **Prior:**

$$\Pr(\phi) = \mathcal{N}(\phi \mid 0, \sigma_p^2 I).$$

15.2 Posterior Distribution

Using Bayes' rule:

$$\Pr(\phi \mid X, w) = \frac{\Pr(w \mid X, \phi) \Pr(\phi)}{\Pr(w \mid X)}.$$

Since the likelihood and prior are not conjugate, the posterior does not have a closed-form solution.

15.3 Laplace Approximation

1. Compute the MAP estimate $\hat{\phi}$ by maximizing:

$$\mathcal{L}(\phi) = \sum_{i=1}^I \left[w_i \log \sigma(a_i) + (1-w_i) \log (1-\sigma(a_i)) \right] - \frac{\phi^\top \phi}{2\sigma_p^2}.$$

2. Approximate the posterior as:

$$\Pr(\phi | X, w) \approx \mathcal{N}(\phi | \hat{\phi}, \Sigma),$$

where

$$\Sigma = -\left(\frac{\partial^2 \mathcal{L}(\phi)}{\partial \phi^2} \Big|_{\phi=\hat{\phi}} \right)^{-1}.$$

15.4 Inference and Predictive Distribution

For a new data point x^* :

$$\Pr(w^* | x^*, X, w) = \int \Pr(w^* | x^*, \phi) \Pr(\phi | X, w) d\phi.$$

A common approximation is:

$$\Pr(w^* = 1 | x^*, X, w) \approx \frac{1}{1 + \exp\left(-\frac{\mu_a}{\sqrt{1+\pi\sigma_a^2/8}}\right)},$$

with

$$\mu_a = \hat{\phi}^\top x^*, \quad \sigma_a^2 = x^{*\top} \Sigma x^*.$$

16 Non-Linear Logistic Regression

16.1 Feature Transformation

Transform the input x into a new feature vector $z = f(x)$.

The model then becomes:

$$\Pr(w = 1 | x, \phi) = \sigma(\phi^\top z).$$

16.2 Examples of Basis Functions

- **Heaviside Step Function:**

$$z_k = \text{Heaviside}[\alpha_k^\top x].$$

- **Arc-Tan Function:**

$$z_k = \arctan(\alpha_k^\top x).$$

- **Radial Basis Function (RBF):**

$$z_k = \exp\left(-\frac{\|x - \alpha_k\|^2}{\lambda_0}\right).$$

16.3 Kernel Logistic Regression

By defining the kernel function

$$K(x_i, x_j) = f(x_i)^\top f(x_j),$$

the model in dual form becomes:

$$\Pr(w = 1 | x, \phi) = \sigma\left(\sum_{i=1}^I \phi_i K(x_i, x)\right).$$

Optimization in dual space relies solely on kernel (dot product) evaluations.

17 Gaussian Process Classification

17.1 GP Classification

Place a prior over functions:

$$f(x) \sim \mathcal{GP}(0, K(x, x')),$$

with $K(x, x')$ the kernel function.

17.2 Predictive Distribution

The predictive probability is:

$$\Pr(w = 1 | x) = \int \sigma(f(x)) \Pr(f | X, w) df.$$

Approximation methods (e.g., Laplace or variational inference) are used since the posterior is non-Gaussian.

18 Incremental Fitting, Boosting, and Trees

18.1 Incremental Fitting

Build a classifier in stages:

$$a_i = \phi_0 + \sum_{k=1}^K \phi_k f[x_i, \xi_k],$$

where each $f[x_i, \xi_k]$ is a (possibly non-linear) feature transformation. Under suitable conditions, this procedure converges to minimize the loss.

18.2 Boosting

Boosting is an ensemble technique that combines weak classifiers. For example, an additive model:

$$a_i = \phi_0 + \sum_{k=1}^K \phi_k \text{Heaviside}[\alpha_k^\top x_i].$$

In AdaBoost, weights are updated as:

$$w_i^{(t+1)} = w_i^{(t)} \exp(-y_i h_t(x_i)),$$

to focus on misclassified examples.

18.3 Branching Logistic Regression

In branching logistic regression, a gating function $g(x, \omega)$ is used to select between two logistic models:

$$a_i = (1 - g(x_i, \omega)) \phi_0^\top x_i + g(x_i, \omega) \phi_1^\top x_i,$$

with $g(x, \omega) \in [0, 1]$, e.g., $g(x, \omega) = \sigma(\omega^\top x)$.

19 Multi-Class Logistic Regression

19.1 Model Formulation

For K classes, use the softmax function:

$$\Pr(w = k | x) = \frac{\exp(a_k)}{\sum_{j=1}^K \exp(a_j)}, \quad a_k = \phi_k^\top x.$$

The softmax guarantees that the probabilities lie in $(0, 1)$ and sum to 1.

19.2 Parameter Learning

The log-likelihood is:

$$L = \sum_{i=1}^I \log \frac{\exp(a_{w_i})}{\sum_{j=1}^K \exp(a_j)},$$

which is equivalent to minimizing the cross-entropy loss. The gradient for class k is:

$$\frac{\partial L}{\partial \phi_k} = \sum_{i=1}^I x_i (y_{ik} - \delta[w_i = k]),$$

where $y_{ik} = \Pr(w_i = k | x_i)$ and $\delta[w_i = k]$ is 1 if $w_i = k$, 0 otherwise.

20 Random Classification Trees

20.1 Binary Tree Splitting

Definition: A random classification tree recursively splits data based on a function $q(x)$ and threshold τ . The log-likelihood for a split at node t is:

$$L = \sum_{i=1}^I \left[\left(1 - \text{heaviside}(q(x_i) - \tau) \right) \log \Pr(w_i | \lambda^{[l]}) + \text{heaviside}(q(x_i) - \tau) \log \Pr(w_i | \lambda^{[r]}) \right]$$

where $\lambda^{[l]}$ and $\lambda^{[r]}$ are the class probabilities for the left and right nodes.

Corollary: For a given τ ,

$$\lambda_k^{[l]} = \frac{\sum_i \delta[w_i = k] (1 - \text{heaviside}[q(x_i) - \tau])}{\sum_i (1 - \text{heaviside}[q(x_i) - \tau])},$$
$$\lambda_k^{[r]} = \frac{\sum_i \delta[w_i = k] \text{heaviside}[q(x_i) - \tau]}{\sum_i \text{heaviside}[q(x_i) - \tau]}.$$

20.2 Related Models

- **Fern:** A simplified tree where all splits at a level use the same function.
- **Random Forest:** An ensemble of trees with predictions averaged:

$$\Pr(w | x) = \frac{1}{T} \sum_{t=1}^T \Pr_t(w | x),$$

where T is the number of trees.

21 Graphical Models

A **graphical model** represents a joint probability distribution using a graph structure. It enables:

- **Factorization:** Decomposing a joint distribution into products of simpler functions.
- **Conditional Independence:** Efficiently encoding independence relationships.

21.1 Conditional Independence

Two variables x_1 and x_3 are conditionally independent given x_2 if

$$\Pr(x_1 | x_2, x_3) = \Pr(x_1 | x_2).$$

If conditional independence holds, the joint density factorizes as:

$$\Pr(x_1, x_2, x_3) = \Pr(x_3 | x_2) \Pr(x_2 | x_1) \Pr(x_1).$$

Graphical models are categorized into:

- **Directed Graphical Models (Bayesian Networks):** Use directed edges to represent causal conditional dependencies.
- **Undirected Graphical Models (Markov Networks):** Use undirected edges to represent symmetric relationships.

22 Directed Graphical Models

22.1 Factorization Rule

A **directed graphical model** factorizes a joint distribution as:

$$\Pr(\mathbf{x}) = \prod_{n=1}^N \Pr(x_n | \text{pa}(n)),$$

where $\text{pa}(n)$ denotes the parents of node x_n .

22.2 Markov Blanket

The **Markov blanket** of a node x_n consists of its parents, children, and the parents of its children. Given its Markov blanket, x_n is conditionally independent of all other nodes.

22.3 Sampling: Ancestral Sampling

Ancestral sampling proceeds by:

1. Sampling root nodes (with no parents) from their marginal distributions.
2. Sampling each subsequent node x_n conditioned on its parents $\text{pa}(n)$ in topological order.

23 Undirected Graphical Models

23.1 Factorization

An **undirected graphical model** expresses the joint distribution as:

$$\Pr(\mathbf{x}) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \phi_C(\mathbf{x}_C),$$

where:

- $\mathcal{C}$ is the set of cliques in the graph.
- ϕ_C is a non-negative potential function defined over clique C .
- Z is the partition function ensuring that the distribution is normalized.

Alternatively, the joint distribution can be written in Gibbs form:

$$\Pr(\mathbf{x}) = \frac{1}{Z} \exp\left(-\sum_{c=1}^C \psi_c(\mathbf{x}_C)\right),$$

where $\psi_c(\mathbf{x}_C) = -\log \phi_c(\mathbf{x}_C)$.

23.2 Sampling: MCMC and Gibbs Sampling

Sampling from an undirected model is often done using Markov Chain Monte Carlo (MCMC) methods such as Gibbs sampling, where:

$$\Pr(x_n | \mathbf{x}_{-n})$$

is used to iteratively update each variable.

24 Graphical Models in Inference

24.1 Inference Techniques

Common inference strategies include:

- **MAP Estimation:** Find $\hat{w} = \arg \max_w \Pr(w | x)$.
- **Marginalization:** Compute $\Pr(w_n | x)$ by summing out other variables.
- **Sampling:** Approximate distributions via methods such as ancestral sampling (directed) or Gibbs sampling (undirected).
- **Directed Models:** Learning involves maximizing the likelihood factorized over nodes:

$$\prod_{i=1}^I \prod_{n=1}^N \Pr(x_{i,n} | \text{pa}(n), \theta).$$

- **Undirected Models:** Learning is based on the log-likelihood:

$$\mathcal{L}(\theta) = \sum_{i=1}^I \sum_{c=1}^C \psi_c(\mathbf{x}_i, \theta) - I \log Z(\theta),$$

which is challenging due to the intractability of Z . Approximation methods, such as contrastive divergence, are often used.

25 Chain and Tree Models

25.1 Chain Models

A **chain model** (e.g., Hidden Markov Model) assumes sequential dependency:

$$\Pr(\mathbf{w}) = \Pr(w_1) \prod_{n=2}^N \Pr(w_n | w_{n-1}),$$

with observations satisfying:

$$\Pr(\mathbf{x} | \mathbf{w}) = \prod_{n=1}^N \Pr(x_n | w_n).$$

25.2 Tree Models

A **tree model** extends chain models to branching structures:

$$\Pr(\mathbf{w}) = \prod_{n=1}^N \Pr(w_n | \text{pa}(n)),$$

where every node (except the root) has one parent.

25.3 MAP Inference in Chains

Define:

$$U_n(w_n) = -\log \Pr(x_n | w_n), \quad P(w_n, w_{n-1}) = -\log \Pr(w_n | w_{n-1}).$$

The cost of a configuration is:

$$\text{Cost}(\mathbf{w}) = \sum_{n=1}^N U_n(w_n) + \sum_{n=2}^N P(w_n, w_{n-1}).$$

Forward Pass: for each node n , compute:

$$S_{n,k} = U_n(w_n = k) + \min_l \left[S_{n-1,l} + P(w_n = k, w_{n-1} = l) \right].$$

25.4 MAP Inference in Trees

Belief propagation is used for MAP inference in tree models. It involves:

- A **forward (upward)** pass: Messages propagate from leaves to the root.
- A **backward (downward)** pass: The optimum is traced back from the root.

26 Marginal Posterior Inference in Chains and Trees

26.1 Forward-Backward

Forward Messages:

$$f_n(w_n) = \Pr(x_n | w_n) \sum_{w_{n-1}} \Pr(w_n | w_{n-1}) f_{n-1}(w_{n-1}),$$

with $f_1(w_1) = \Pr(x_1 | w_1) \Pr(w_1)$.

Backward Messages: Similarly, define messages from $n = N$ back.

$$\Pr(w_n | \mathbf{x}) \propto f_n(w_n) b_n(w_n).$$

26.2 Sum-Product Algorithm

The sum-product algorithm generalizes forward-backward to trees:

- **Messages are passed** between variable and factor nodes.
- The marginal at node w_n is computed by multiplying all incoming messages and normalizing.

27 Models with Loops

In loopy graphs (e.g., grid graphs in vision), cycles hinder the use of exact dynamic programming due to overlapping subproblems.

27.1 Inference for Loops

- **Loopy Belief Propagation:** Iteratively update messages in graphs with cycles. Convergence is not guaranteed but often works well.

- **Graph Pruning:** Remove some edges to form a tree (at the cost of losing dependencies).
- **Variable Combination:** Merge nodes to form a tree structure (can be computationally expensive).
- **Sampling:** Use Monte Carlo methods to approximate marginals.
- **Specialized Techniques:** Tree-reweighted message passing or graph cuts.

27.2 Loopy Belief Propagation

Loopy belief propagation generalizes belief propagation to graphs with cycles by iteratively updating messages:

$$m_{x \rightarrow g} = \prod_{f \in \text{ne}(x) \setminus g} m_{f \rightarrow x},$$
$$m_{g \rightarrow x} = \sum_{\mathbf{z}} \left[g(\mathbf{z}) \prod_{y \in \text{ne}(g) \setminus x} m_{y \rightarrow g} \right].$$